

Chapter 3. **GROUPS**

3.3 Cyclic Groups

3.3.2 The Fundamental Theorem of Finite Cyclic Groups



In this video we will prove the

Fundamental Theorem of Finite Cyclic Groups

Proving the Fundamental Theorem of Finite cyclic groups

THEOREM 3.30. FUNDAMENTAL THEOREM OF FINITE CYCLIC GROUPS.

Let $G = \langle g \rangle$ be a cyclic group of order n .

- (1) If H is a subgroup of G , then $H = \langle g^d \rangle$ for some $d|n$.
- (2) If H is any subgroup of G with order $|H| = k$, then $k|n$.
- (3) If $k|n$, then $\langle g^{n/k} \rangle$ is the unique subgroup of G of order k .

LEMMA 3.31. Let g be an element of a group G of order n . If $d|n$ then the order of the element g^d is n/d .

Proof Suppose that $G = \langle g \rangle$ is a cyclic group of order n .

① Take H a subgroup of G . If $|H| = 1$, then there is nothing to prove. Otherwise, we know that H is a cyclic group, say that

$H = \langle g^m \rangle$. Let $d = \gcd(m, n)$.

We claim that $H = \langle g^d \rangle$. Notice that $d|n$.

$$\langle g^m \rangle \stackrel{?}{=} \langle g^d \rangle$$

It is enough to prove that $g^m \in \langle g^d \rangle$

and $g^d \in \langle g^m \rangle$.

From $d \mid m$, we know that $m = dk$ for some integer k . But then

$$\underline{g^m = g^{dk} = (g^d)^k \in \langle g^d \rangle}$$

On the other hand, by Bézout's Lemma we have that $d = xm + yn$ for some integers x and y .

$$\begin{aligned} g^d &= g^{xm+yn} = g^{xm} \cdot g^{yn} = (g^m)^x \cdot (g^n)^y \stackrel{g^n=1}{=} \\ &= (g^m)^x \in \langle g^m \rangle \end{aligned}$$

This finishes the proof of (4).

② Suppose that H is a subgroup of G of order k . From ① we know that

$$H = \langle g^d \rangle, \text{ with } d \mid n.$$

$$k = |H| = o(g^d) \stackrel{\text{Lemma}}{=} \frac{n}{d} \Rightarrow n = kd \Rightarrow \underline{k \mid n}$$

③ Suppose that H is a subgroup of G of order k . By ① we have that

$$H = \langle g^m \rangle \text{ with } m \mid n. \text{ But then}$$

by the previous lemma we have that

$$k = |H| = o(g^m) = \frac{n}{m} \Rightarrow m = \frac{n}{k}$$

$$\text{and so } H = \langle g^{\frac{n}{k}} \rangle.$$



COROLLARY 3.32. Let $G = \langle g \rangle$ be a cyclic group of order n . Then for any positive integer m we have that $\langle g^m \rangle = \langle g^d \rangle$, where $d = \gcd(m, n)$. Moreover, the order of g^m is n/d .

Proof. Take m any positive integer and let $d = \gcd(m, n)$. Proceeding like in (1) in the previous proof we have that

$$\langle g^m \rangle = \langle g^d \rangle.$$

On the other hand

$$o(g^m) = |\langle g^m \rangle| = |\langle g^d \rangle| = o(g^d)$$

$$\stackrel{\text{lemma}}{=} \frac{n}{d}$$

